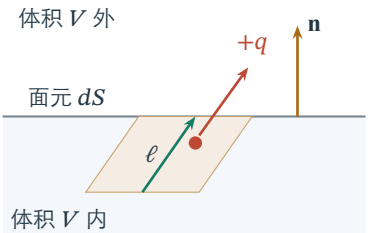
理论图 T16　极化通量与闭合体积内留下的电荷



只有贴近边界且能穿出面元的正电荷，才贡献穿出通量。

$$dV_{\text{col}} = \ell \cdot d\mathbf{S}$$

$$\begin{aligned} dQ_{\text{out}} &= Nq\ell \cdot d\mathbf{S} \\ &= \mathbf{P} \cdot d\mathbf{S} \end{aligned}$$

$$\begin{aligned} Q_b &= -Q_{\text{out}} \\ &= -\oiint_S \mathbf{P} \cdot d\mathbf{S} \end{aligned}$$